

Stagehand Phase Evidence

Current checkpoint: public Site v89, verified 2026-09-03.

Project

Stagehand is a WebMCP-native 2D animation studio for editable paper-cutout scenes. The current cut supports rigged characters, imported image props, per-asset visual direction, camera work, captions, non-voice audio cues, storyboard beats, multi-scene Preview, and deterministic WebM export.

Private source: github.com/swaymun/stagehand-animation-studio

Commit: `caa365248f276aaeaca069c0498c2f03059a44bc`

Public Site: stagehand-animation-studio.saimun-h-shahee.chatgpt.site

Sites version: 89

Acceptance loop

Implement - local tests - Sites deployment - hosted test - UI roast - fixes - second review.

Implemented at this checkpoint

- Revision-safe WebMCP mutations with optional `expectedRevision` and structured `REVISION_CONFLICT` results.
- Idempotent WebMCP mutations with optional `idempotencyKey` replay protection.
- Structured per-asset role, treatment, silhouette, palette, and direction notes with human and agent parity.
- Asset treatment choices reach canvas, thumbnails, Preview, and WebM; new assets also receive valid style defaults immediately.
- Motion actions show affected character and bounded duration; pose presets remain a separate group.
- Human audio cue Start/End editors share bounded timing updates with the agent cue editor.
- Storyboard mode selects the Board rail; Preview hides editing mutation surfaces and selection chrome.

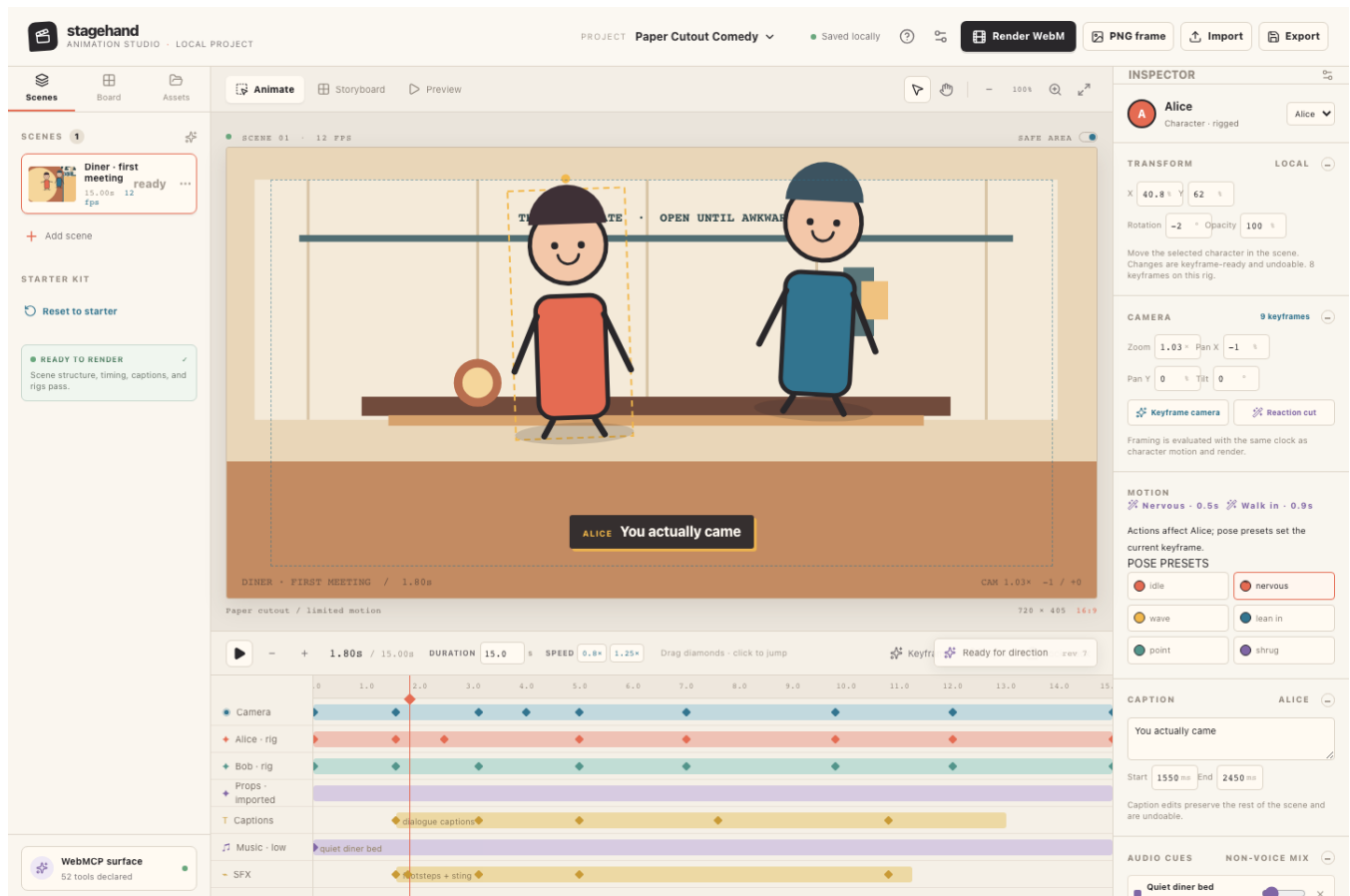
Verification

Local gate:

```
npm run format
npm run lint
npm run build
git diff --check
npm run smoke
npm run smoke:native
```

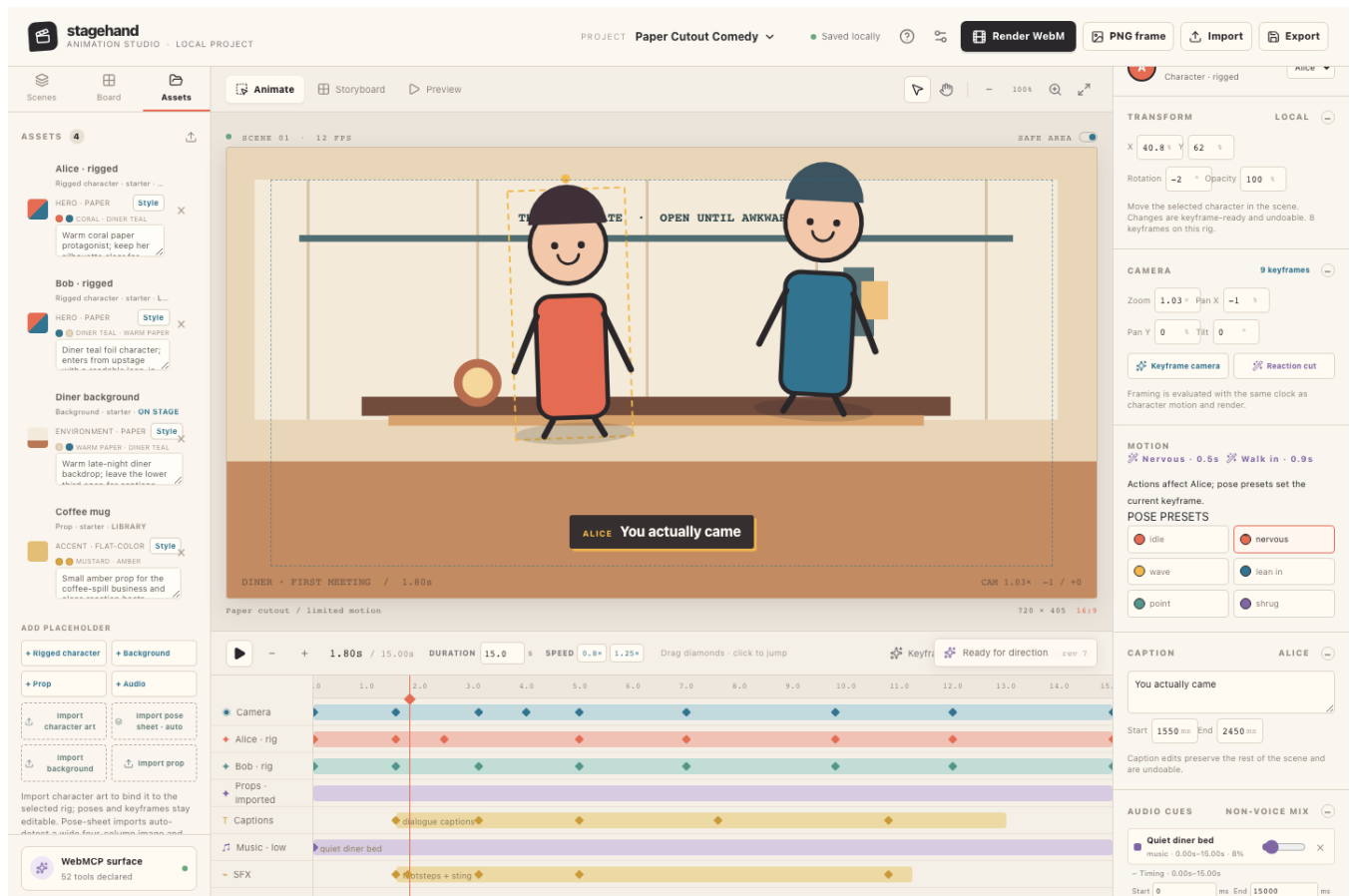
Smoke passed: 52 tools, 39 guarded mutations, a 15-second six-beat starter, synchronized timing, asset treatments, import/recovery including empty collections, Preview, valid PNG/WebM signatures, native WebMCP registration, and zero errors.

Animate editing workspace



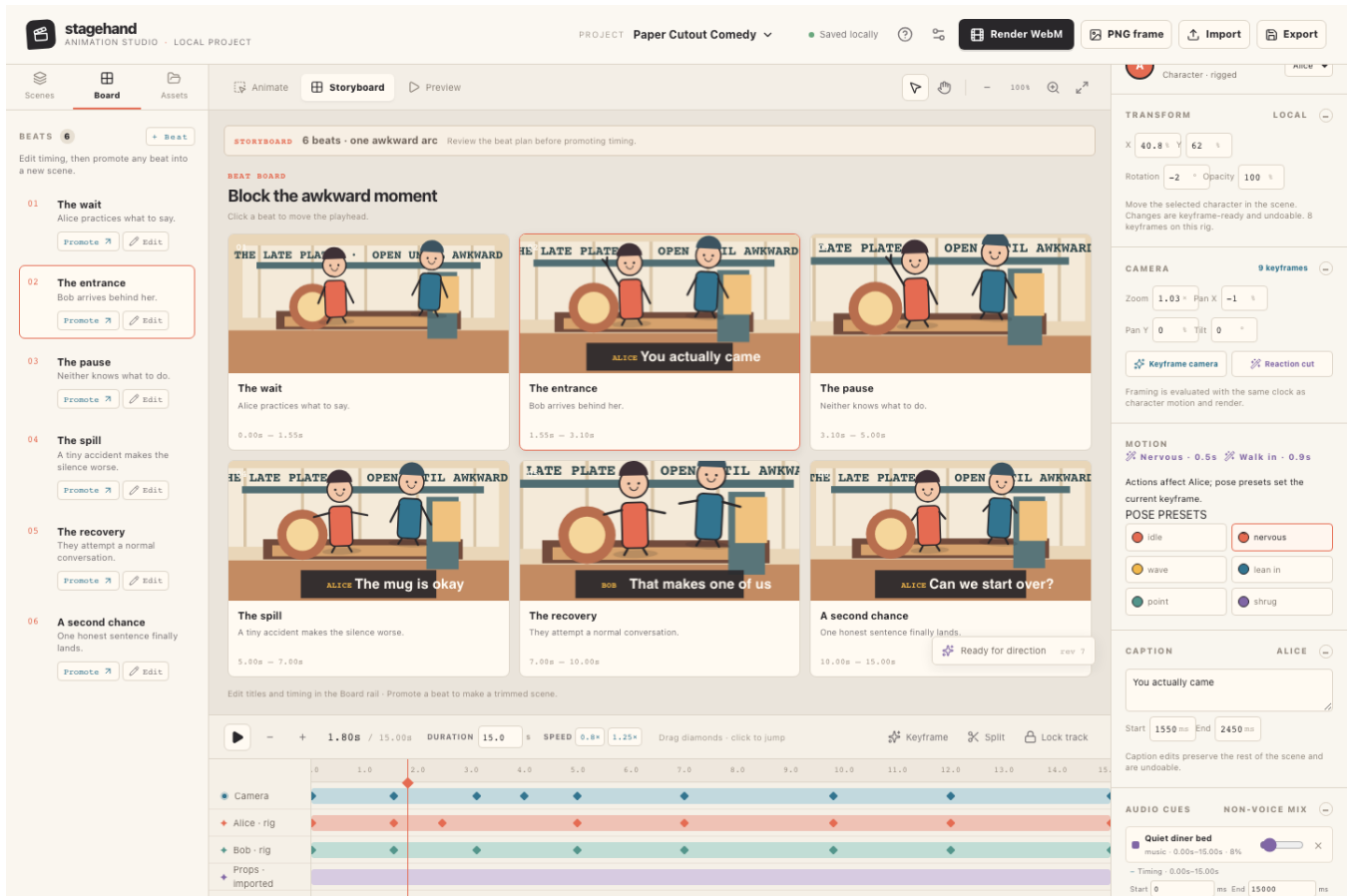
Hosted Playwright capture at 1440 x 960. The 15-second timeline carries the expanded character and camera beats while speed controls, frame steps, and labels remain readable; v89 has no visual surface change.

Assets style editor



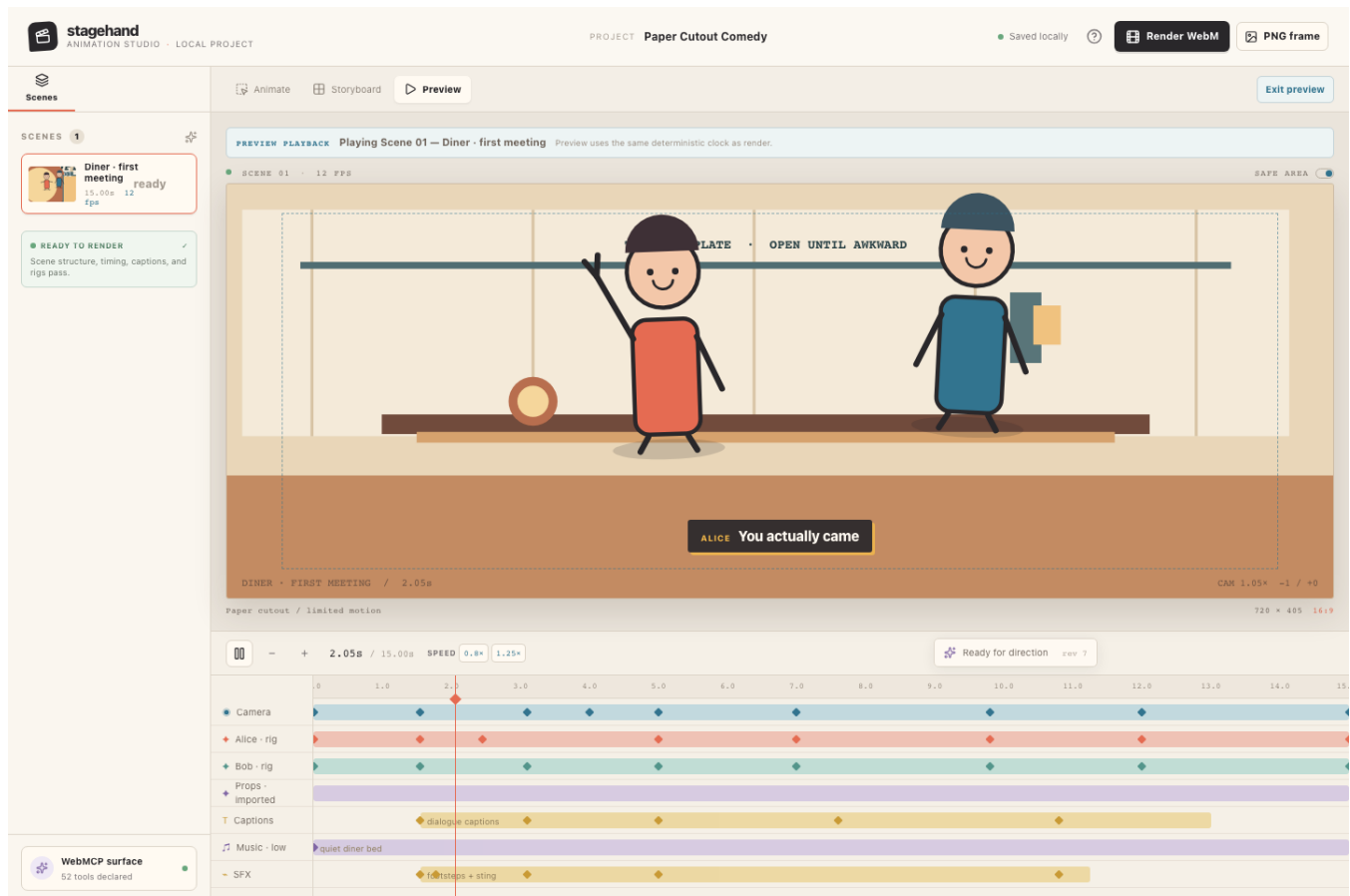
Hosted Playwright capture at 1440 x 960. Palette chips and stage/library cues make the structured Style direction legible without taking over the asset rail.

Storyboard mode with Board rail



Hosted Playwright capture at 1440 x 960. Six beat cards make the longer arc inspectable while the selected mode and project rail stay synchronized.

Review-first Preview player



Hosted Playwright capture at 1440 x 960. Scene-only context, transport, and Exit preview remain; global editing controls are hidden.

Roast findings and fixes

- Imported props lacked direct human transform editing. Added X/Y/Scale/Rot controls sharing the keyframe command path.
- Four prop transform fields initially collided with the asset-list flex rule and truncated. Replaced it with an explicit two-column grid; computed field boxes are 57 px wide in two rows.
- Preview carried too much editing chrome. Converted it into a review-first player with scene-only context and Exit preview.
- Timeline diamonds had 7 x 7 px hit targets. Enlarged the buttons to 22 x 22 pixels while preserving the visual diamond.
- Agent retries could duplicate successful writes. Added optional idempotency-key replay protection and a hosted smoke assertion.
- Switching top-level Storyboard mode could leave Assets selected. Synced mode selection to its companion project rail.
- Character transform inputs lacked explicit accessible names. Added dynamic X/Y/rotation/opacity labels and a smoke check.
- Asset briefs carried style intent only as prose. Added structured role, treatment, silhouette, palette, and direction fields; later made those treatments visible in the canvas and manifest placement cues.
- Motion action buttons were visually mixed with pose presets. Added explicit durations, affected-character copy, and a Pose presets label.
- Agent frame-rate-only render updates reset an existing 1080p choice. Preserved the current resolution when the preset is omitted and added smoke coverage.
- Preview still exposed global project-editing controls. Hid Settings, Import, Export JSON, and rename while retaining WebM and PNG review actions.
- Scene titles competed with the ready badge and were visually shortened. Used a two-line title clamp and positioned the badge outside the text flow.
- The smoke gate did not exercise recovery, modal semantics, or cross-scene playback. Added Help open/close, reload recovery, and sequence Preview transition checks to local and hosted runs.
- The Inspector was a long scroll dump, human scrubbing could leave agent reads one event behind, and keyboard stepping was unreliable after scrub focus. Added native collapsible groups, shared project-view synchronization, a focused stage target, and smoke coverage.
- Export checks only proved that downloads were non-empty. Added binary signature assertions for WebM/EBML and PNG outputs in local and hosted smoke runs.
- Experimental Chromium exposed navigator.modelContext while the app only checked document.modelContext. Added a compatibility fallback and a native registration smoke gate covering all 52 tools plus valid and invalid calls.

Limits

Direct CUA inspection was unavailable because the Mac was locked; Playwright was used for hosted interaction and screenshots. Native Chromium smoke proves browser registration and callback behavior, while ChatGPT's in-app browser discovery remains unverified. TTS, voice cloning, dialogue recording, phoneme extraction, and lip-sync remain out of scope.